

Hybrid Anomaly Detection Model: Post-Mortem and Path Forward

Multivariate time-series anomaly detection — CNN Autoencoder + Isolation Forest

1. Executive Summary

The hybrid model (a CNN autoencoder fused with a window-based Isolation Forest) was built to detect anomalies in multivariate sensor time-series data using an unsupervised approach, since training data carries no anomaly labels. After two rounds of bug fixes and tuning, both branches plateaued at a modest ROC-AUC of 0.61-0.65 and a validation F1 in the 0.45-0.49 range, confirmed by real competition-portal scores (public F1 0.399, private F1 0.471). Every threshold, fusion, and hyperparameter bug we could find has been fixed; the score has not moved meaningfully because the underlying signal from both branches is capped, not because of a calibration bug. This report documents that root-cause diagnosis and the reasoning for moving next to a dilated CNN (TCN-style) architecture.

2. Background: What the Hybrid Model Does

Each raw file is one independent sensor-recording run. The hybrid pipeline: (1) scales each run with a StandardScaler fit on train only, (2) slices runs into 200-timestep sliding windows, (3) trains a CNN autoencoder to reconstruct normal windows (unsupervised - reconstruction error becomes the anomaly score), (4) trains an Isolation Forest on 7 summary statistics per sensor per window (statistical outlier score), (5) normalizes and fuses both scores, and (6) selects a threshold that maximizes F1 on the small labeled validation set (10 runs). Test data is unlabeled and scored externally by the competition portal.

3. Work Performed Across Two Rounds

Round 1 - repository cleanup and correctness fixes:

- Isolation Forest window features were flattening all 18 sensors into a single scalar per statistic (no axis=0) - fixed to compute per-sensor statistics.
- Threshold selection had an off-by-one against precision_recall_curve's output, so the applied threshold never matched the reported best F1 - fixed.
- Windows were built by concatenating every file in a split before slicing, splicing one run's tail onto another run's head - fixed to window per run.
- Validation/test were scored at one label per 10-timestep block; rebuilt to score densely (stride 1) and average overlapping windows per raw timestep, matching the true submission granularity.
- CNN epoch count, silently reverted to 10 in the production pipeline despite the project's own tuning notebook proving 50 was needed, was restored.
- Fusion weight (previously hardcoded 0.85/0.15) is now swept against validation.

Round 2 - targeted model-quality improvements:

- Added validation-based checkpoint selection to CNN training (evaluate every 5 epochs, keep the best-scoring checkpoint) after observing the reconstruction loss fall smoothly for all 50 epochs with no plateau - a classic overtraining signature.
- Swept Isolation Forest's contamination parameter (0.1-0.3) against validation.
- Widened the fusion weight sweep grid from four coarse candidates to nine (0.1 steps).

4. Results

Model / Stage	F1	Precision	Recall	ROC-AUC
CNN (quoted baseline, window-level)	0.60	-	-	-
Isolation Forest (quoted baseline, window-level)	0.40	-	-	-
Hybrid (quoted baseline, window-level)	0.62	-	-	-
CNN standalone - Round 1 (per-timestep, val)	0.408	0.298	0.644	0.574
Isolation Forest window - Round 1 (per-timestep, val)	0.491	0.334	0.922	0.649
Hybrid - Round 1 (per-timestep, val)	0.494	0.337	0.921	0.636
CNN standalone - Round 2 (best checkpoint, val)	0.458	0.324	0.777	0.614
Isolation Forest window - Round 2 (val)	0.491	0.334	0.922	0.649
Hybrid - Round 2 (val)	0.494	0.337	0.924	0.647
Hybrid - real competition portal (public)	0.399	-	-	-
Hybrid - real competition portal (private)	0.471	-	-	-

Note: baseline rows were reported at window-level granularity (one label per ~200-timestep block); all other rows use the corrected per-timestep granularity, which is stricter and matches the real submission format, so the two groups are not directly comparable. Portal scores are the only true test-set numbers; everything else is validation-set (test is unlabeled locally).

Two findings stand out. First, Round 2's checkpoint selection produced a real, reproducible gain for the CNN branch alone (+0.05 F1, found by observing validation F1 peak at epoch 5 rather than epoch 50). Second, and more importantly, that gain barely moved the hybrid score (0.494 to 0.494) - improving one branch's standalone accuracy did not make it a materially better ensemble partner, and the fusion weight sweep shifted further toward Isolation Forest (0.4 CNN / 0.6 ISO) rather than staying balanced.

5. Root Cause Analysis: Why the Hybrid Model Underperformed

5.1 Weak score separability, not a calibration bug

ROC-AUC for both branches sits at 0.61-0.65 (0.5 is random, 1.0 is perfect separation). Every threshold and calibration bug identified in this project has already been fixed - the off-by-one in threshold selection, the hardcoded fusion weight, the hardcoded Isolation Forest contamination (confirmed via a direct sweep to have zero effect on the score, since sklearn's contamination parameter only shifts the model's own internal classification cutoff, not the continuous decision_function output this pipeline actually scores with). F1 has stayed in the same 0.40-0.49 band across all of these fixes. That pattern is the signature of a signal ceiling in the underlying model/features, not an engineering bug still left to find.

5.2 The CNN autoencoder is architecturally too shallow for the anomaly pattern

The project's own tuning experiments (window-size sweep) already established that anomalies in this dataset are slow, sustained drifts rather than sharp spikes - F1 improved monotonically as window size increased from 20 to 200 timesteps. The current architecture is a single shallow Conv1d encoder/decoder block (32 to 64 channels, one pooling layer). Its effective receptive field is small relative to the 200-step window it is given, meaning it can see the window but cannot necessarily model long-range dependencies within it as effectively as a deeper or dilated architecture could.

5.3 Likely training-set contamination

The autoencoder is trained purely unsupervised on the train split, which carries no anomaly labels and is not guaranteed to be anomaly-free. During the Round 2 retraining run, validation F1 peaked early (epoch 5, F1 0.458) and then oscillated downward with no recovery through epoch 50 (ending at F1 0.409), while training reconstruction loss fell smoothly the entire time with no plateau. This is consistent with the model gradually learning to reconstruct some anomalous patterns present in the unlabeled training data, which directly shrinks the normal-versus-anomalous reconstruction-error gap the anomaly score depends on.

5.4 Isolation Forest's features are informative but shallow

The window-based Isolation Forest branch used in the hybrid represents each window as 7 summary statistics per sensor (mean, std, min, max, median, and two percentiles) - a real improvement over the original bug (which flattened all sensors into one scalar), but still a purely static, non-temporal representation. It has no cross-sensor interaction terms and no frequency-domain features, unlike the separate tabular Isolation Forest baseline elsewhere in this project, which does include interaction features.

5.5 The fusion is not adding the value it should

The premise of the hybrid design is that the CNN and Isolation Forest fail on different kinds of anomalies, so combining them should cover more ground than either alone. In practice, the fused hybrid score (F1 0.494) sits barely above the Isolation Forest branch alone (F1 0.491), and improving the CNN branch's own accuracy in Round 2 did not translate into a better hybrid score. This suggests the two branches' errors are more correlated than the design assumed, rather than being complementary.

5.6 Systematic over-prediction of anomalies

At the F1-optimal threshold, the model predicts roughly 65-88% of timesteps as anomalous, against a true validation anomaly rate of 24%. Per test run, 39 to 40 of 53 runs are predicted as entirely anomalous end to end. This is a direct consequence of weak score separability (5.1-5.5): when normal and anomalous windows are not well separated, the threshold that happens to maximize F1 buys very high recall at the cost of precision, and the practical result is a model that over-flags at a rate far beyond what is plausible for a real anomaly-detection task.

6. Why Move to a Dilated CNN (TCN-style) Architecture Next

Of the root causes above, 5.2 (architectural depth) is the one most directly actionable without changing the project's overall unsupervised-autoencoder approach, and the project's own data already points to the fix: since wider windows already improved every result recorded, an architecture with a larger effective receptive field should help the model actually use the temporal context it is given, rather than only being handed a wider window.

Three architecture families were considered:

- **LSTM/GRU autoencoder** - naturally models long-range sequential dependencies, but is typically slower to train, less stable on 200-step sequences, and does not parallelize on GPU as efficiently per epoch as a convolutional approach.
- **Transformer-based autoencoder** (e.g. Anomaly Transformer style) - captures long-range dependencies very well via attention and represents a modern approach in the time-series anomaly detection literature, but is data-hungry and adds meaningfully more tuning risk and complexity than is justified given this project's data volume and timeline.
- **Dilated CNN (TCN-style) autoencoder** - exponentially expands the receptive field by stacking dilated convolutions, without enlarging the window or increasing memory cost, and is a close, low-risk extension of the existing ConvAutoencoder1D already in the codebase.

The dilated CNN is the recommended next step: it directly targets the empirically-confirmed need for wider effective temporal context, reuses the existing training/scoring/evaluation pipeline almost unchanged, and carries the lowest implementation and training-stability risk of the three options. It is expected to improve the CNN branch's standalone separability meaningfully; based on the gap between current ROC-AUC (~0.61) and a well-separated model (~0.80+), a realistic target for this round is incremental improvement into the F1 0.55-0.65 range rather than a large jump, given the unsupervised training setup and limited labeled validation data remain unchanged.

7. Recommended Next Steps

- Replace the single-block Conv1d encoder/decoder with a stacked dilated causal convolution architecture (TCN-style), reusing the existing SensorDataset, training loop, and per-timestep evaluation pipeline.
- Re-use the Round 2 checkpoint-selection mechanism during training, since overtraining (Section 5.3) is independent of architecture and will still need to be guarded against.

- Separately investigate contamination-robust training (down-weighting high-reconstruction-error windows during training) as a complementary fix to 5.3, since it addresses a different failure mode than architecture depth.
- Re-run the fusion weight sweep once the new CNN branch is retrained, since Section 5.5 shows the optimal fusion point shifts whenever branch quality changes.